# Section A: Pure Mathematics

**1** *You need not consider the convergence of the improper integrals in this question.*

For $p, q > 0$, define

$$\mathrm{b}(p,q) = \int_0^1 x^{p-1}(1-x)^{q-1}\,\mathrm{d}x.$$

**(i)** Show that $\mathrm{b}(p,q) = \mathrm{b}(q,p)$.

**(ii)** Show that $\mathrm{b}(p+1,q) = \mathrm{b}(p,q) - \mathrm{b}(p,q+1)$ and hence that $\mathrm{b}(p+1,p) = \frac{1}{2}\,\mathrm{b}(p,p)$.

**(iii)** Show that

$$\mathrm{b}(p,q) = 2\int_0^{\frac{1}{2}\pi} (\sin\theta)^{2p-1}(\cos\theta)^{2q-1}\,\mathrm{d}\theta.$$

Hence show that $\mathrm{b}(p,p) = \dfrac{1}{2^{2p-1}}\,\mathrm{b}\big(p,\tfrac{1}{2}\big)$.

**(iv)** Show that

$$\mathrm{b}(p,q) = \int_0^\infty \frac{t^{p-1}}{(1+t)^{p+q}}\,\mathrm{d}t.$$

**(v)** Evaluate

$$\int_0^\infty \frac{t^{\frac{3}{2}}}{(1+t)^6}\,\mathrm{d}t.$$

**2** Let $\mathrm{f}(x) = 7 - 2|x|$.

A sequence $u_0, u_1, u_2, \ldots$ is defined by $u_0 = a$ and $u_n = \mathrm{f}(u_{n-1})$ for $n > 0$.

**(i)** **(a)** Sketch, on the same axes, the graphs with equations $y = \mathrm{f}(x)$ and $y = \mathrm{f}(\mathrm{f}(x))$.

**(b)** Find all solutions of the equation $\mathrm{f}(\mathrm{f}(x)) = x$.

**(c)** Find the values of $a$ for which the sequence $u_0, u_1, u_2, \ldots$ has period 2.

**(d)** Show that, if $a = \frac{28}{5}$, then the sequence $u_2, u_3, u_4, \ldots$ has period 2, but neither $u_0$ or $u_1$ is equal to either of $u_2$ or $u_3$.

**(ii)** **(a)** Sketch, on the same axes, the graphs with equations $y = \mathrm{f}(x)$ and $y = \mathrm{f}(\mathrm{f}(\mathrm{f}(x)))$.

**(b)** Consider the sequence $u_0, u_1, u_2, \ldots$ in the cases $a = 1$ and $a = -\frac{7}{9}$. Hence find all the solutions of the equation $\mathrm{f}(\mathrm{f}(\mathrm{f}(x))) = x$.

**(c)** Find a value of $a$ such that the sequence $u_3, u_4, u_5, \ldots$ has period 3, but where none of $u_0$, $u_1$ or $u_2$ is equal to any of $u_3$, $u_4$ or $u_5$.

**3** Let $\mathrm{f}(x)$ be defined and positive for $x > 0$.

Let $a$ and $b$ be real numbers with $0 < a < b$ and define the points $A = (a, \mathrm{f}(a))$ and $B = (b, -\mathrm{f}(b))$.

Let $X = (m, 0)$ be the point of intersection of line $AB$ with the $x$-axis.

**(i)** Find an expression for $m$ in terms of $a$, $b$, $\mathrm{f}(a)$ and $\mathrm{f}(b)$.

**(ii)** Show that, if $\mathrm{f}(x) = \sqrt{x}$, then $m = \sqrt{ab}$.

Find, in terms of $n$, a function $\mathrm{f}(x)$ such that $m = \dfrac{a^{n+1} + b^{n+1}}{a^n + b^n}$.

**(iii)** Let $\mathrm{g}_1(x)$ and $\mathrm{g}_2(x)$ be defined and positive for $x > 0$. Let $m = M_1$ when $\mathrm{f}(x) = \mathrm{g}_1(x)$ and let $m = M_2$ when $\mathrm{f}(x) = \mathrm{g}_2(x)$.

Show that if $\dfrac{\mathrm{g}_1(x)}{\mathrm{g}_2(x)}$ is a decreasing function then $M_1 > M_2$.

Hence show that

$$\frac{a+b}{2} > \sqrt{ab} > \frac{2ab}{a+b}.$$

**(iv)** Let $p$ and $c$ be chosen so that the curve $y = p(c - x)^3$ passes through both $A$ and $B$. Show that

$$\frac{c-a}{b-c} = \left(\frac{\mathrm{f}(a)}{\mathrm{f}(b)}\right)^{\frac{1}{3}}$$

and hence determine $c$ in terms of $a$, $b$, $\mathrm{f}(a)$ and $\mathrm{f}(b)$.

Show that if f is a decreasing function, then $c < m$.

**4** **(i)** $x_2$ and $y_2$ are defined in terms of $x_1$ and $y_1$ by the equation

$$\begin{pmatrix} x_2 \\ y_2 \end{pmatrix} = \begin{pmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix} \begin{pmatrix} x_1 \\ y_1 \end{pmatrix}.$$

$G_1$ is the graph with equation

$$\frac{x^2}{9} + \frac{y^2}{4} = 1$$

and $G_2$ is the graph with equation

$$\frac{\left(\frac{x}{\sqrt{2}} + \frac{y}{\sqrt{2}}\right)^2}{9} + \frac{\left(-\frac{x}{\sqrt{2}} + \frac{y}{\sqrt{2}}\right)^2}{4} = 1.$$

Show that, if $(x_1, y_1)$ is a point on $G_1$, then $(x_2, y_2)$ is a point on $G_2$.

Show that $G_2$ is an anti-clockwise rotation of $G_1$ through $45°$ about the origin.

**(ii)** **(a)** The matrix

$$\begin{pmatrix} -0.6 & 0.8 \\ 0.8 & 0.6 \end{pmatrix}$$

represents a reflection. Find the line of invariant points of this matrix.

**(b)** Sketch, on the same axes, the graphs with equations

$$y = 2^x \text{ and } 0.8x + 0.6y = 2^{-0.6x+0.8y}.$$

**(iii)** Sketch, on the same axes, for $0 \leqslant x \leqslant 2\pi$, the graphs with equations

$$y = \sin x \text{ and } y = \sin(x - 2y).$$

You should determine the exact co-ordinates of the points on the graph with equation $y = \sin(x - 2y)$ where the tangent is horizontal and those where it is vertical.

**5** Three points, $A$, $B$ and $C$, lie in a horizontal plane, but are not collinear. The point $O$ lies above the plane.

Let $\overrightarrow{OA} = \mathbf{a}$, $\overrightarrow{OB} = \mathbf{b}$ and $\overrightarrow{OC} = \mathbf{c}$.

$P$ is a point with $\overrightarrow{OP} = \alpha\mathbf{a} + \beta\mathbf{b} + \gamma\mathbf{c}$, where $\alpha$, $\beta$ and $\gamma$ are all positive and $\alpha + \beta + \gamma < 1$.

Let $k = 1 - (\alpha + \beta + \gamma)$.

**(i)** The point $L$ is on $OA$, the point $X$ is on $BC$ and $LX$ passes through $P$.

Determine $\overrightarrow{OX}$ in terms of $\beta$, $\gamma$, $\mathbf{b}$ and $\mathbf{c}$ and show that $\overrightarrow{OL} = \dfrac{\alpha}{k+\alpha}\mathbf{a}$.

**(ii)** Let $M$ and $Y$ be the unique pair of points on $OB$ and $CA$ respectively such that $MY$ passes through $P$, and let $N$ and $Z$ be the unique pair of points on $OC$ and $AB$ respectively such that $NZ$ passes through $P$.

Show that the plane $LMN$ is also horizontal if and only if $OP$ intersects plane $ABC$ at the point $G$, where $\overrightarrow{OG} \equiv \frac{1}{3}(\mathbf{a} + \mathbf{b} + \mathbf{c})$. Where do points $X$, $Y$ and $Z$ lie in this case?

**(iii)** State what the condition $\alpha + \beta + \gamma < 1$ tells you about the position of $P$ relative to the tetrahedron $OABC$.

**6** **(i)** Let $a$, $b$ and $c$ be three non-zero complex numbers with the properties $a + b + c = 0$ and $a^2 + b^2 + c^2 = 0$.

Show that $a$, $b$ and $c$ cannot all be real.

Show further that $a$, $b$ and $c$ all have the same modulus.

**(ii)** Show that it is not possible to find three non-zero complex numbers $a$, $b$ and $c$ with the properties $a + b + c = 0$ and $a^3 + b^3 + c^3 = 0$.

**(iii)** Show that if any four non-zero complex numbers $a$, $b$, $c$ and $d$ have the properties $a + b + c + d = 0$ and $a^3 + b^3 + c^3 + d^3 = 0$, then at least two of them must have the same modulus.

**(iv)** Show, by taking $c = 1$, $d = -2$ and $e = 3$ that it is possible to find five real numbers $a$, $b$, $c$, $d$ and $e$ with distinct magnitudes and with the properties $a + b + c + d + e = 0$ and $a^3 + b^3 + c^3 + d^3 + e^3 = 0$.

**7** Let $\mathrm{f}(x) = \sqrt{x^2+1} - x$.

**(i)** Using a binomial series, or otherwise, show that, for large $|x|$, $\sqrt{x^2+1} \approx |x| + \dfrac{1}{2|x|}$.

Sketch the graph $y = \mathrm{f}(x)$.

**(ii)** Let $\mathrm{g}(x) = \tan^{-1}\mathrm{f}(x)$ and, for $x \neq 0$, let $\mathrm{k}(x) = \frac{1}{2}\tan^{-1}\dfrac{1}{x}$.

**(a)** Show that $\mathrm{g}(x) + \mathrm{g}(-x) = \frac{1}{2}\pi$.

**(b)** Show that $\mathrm{k}(x) + \mathrm{k}(-x) = 0$.

**(c)** Show that $\tan \mathrm{k}(x) = \tan \mathrm{g}(x)$ for $x > 0$.

**(d)** Sketch the graphs $y = \mathrm{g}(x)$ and $y = \mathrm{k}(x)$ on the same axes.

**(e)** Evaluate $\displaystyle\int_0^1 \mathrm{k}(x)\,\mathrm{d}x$ and hence write down the value of $\displaystyle\int_{-1}^0 \mathrm{g}(x)\,\mathrm{d}x$.

**8** **(i)** Show that

$$z^{m+1} - \frac{1}{z^{m+1}} = \left(z - \frac{1}{z}\right)\left(z^m + \frac{1}{z^m}\right) + \left(z^{m-1} - \frac{1}{z^{m-1}}\right).$$

Hence prove by induction that, for $n \geqslant 1$,

$$z^{2n} - \frac{1}{z^{2n}} = \left(z - \frac{1}{z}\right)\sum_{r=1}^{n}\left(z^{2r-1} + \frac{1}{z^{2r-1}}\right).$$

Find similarly $z^{2n} - \dfrac{1}{z^{2n}}$ as a product of $\left(z + \dfrac{1}{z}\right)$ and a sum.

**(ii)** **(a)** By choosing $z = \mathrm{e}^{\mathrm{i}\theta}$, show that

$$\sin 2n\theta = 2\sin\theta \sum_{r=1}^{n} \cos(2r-1)\theta\,.$$

**(b)** Use this result, with $n = 2$, to show that $\cos\frac{2}{5}\pi = \cos\frac{1}{5}\pi - \frac{1}{2}$.

**(c)** Use this result, with $n = 7$, to show that $\cos\frac{2}{15}\pi + \cos\frac{4}{15}\pi + \cos\frac{8}{15}\pi + \cos\frac{16}{15}\pi = \frac{1}{2}$.

**(iii)** Show that $\sin\frac{1}{14}\pi - \sin\frac{3}{14}\pi + \sin\frac{5}{14}\pi = \frac{1}{2}$.

## Section B: Mechanics

**9** In this question, $n \geqslant 2$.

**(i)** A solid, of uniform density, is formed by rotating through 360° about the $y$-axis the region bounded by the part of the curve $r^{n-1}y = r^n - x^n$ with $0 \leqslant x \leqslant r$, and the $x$- and $y$-axes.

Show that the $y$-coordinate of the centre of mass of this solid is $\dfrac{nr}{2(n+1)}$.

**(ii)** Show that the normal to the curve $r^{n-1}y = r^n - x^n$ at the point $\big(rp, r(1-p^n)\big)$, where $0 < p \leqslant 1$, meets the $y$-axis at $(0, Y)$, where $Y = r\left(1 - p^n - \dfrac{1}{np^{n-2}}\right)$.

In the case $n = 4$, show that the greatest value of $Y$ is $\frac{1}{4}r$.

**(iii)** A solid is formed by rotating through 360° about the $y$-axis the region bounded by the curves $r^3y = r^4 - x^4$ and $ry = -(r^2 - x^2)$, both for $0 \leqslant x \leqslant r$.

$A$ and $B$ are the points $(0, -r)$ and $(0, r)$, respectively, on the surface of the solid.

Show that the solid can rest in equilibrium on a horizontal surface with the vector $\overrightarrow{AB}$ at three different, non-zero, angles to the upward vertical. You should not attempt to find these angles.

**10** A plank $AB$ of length $L$ initially lies horizontally at rest along the $x$-axis on a flat surface, with $A$ at the origin.

Point $C$ on the plank is such that $AC$ has length $sL$, where $0 < s < 1$.

End $A$ is then raised vertically along the $y$-axis so that its height above the horizontal surface at time $t$ is $\mathrm{h}(t)$, while end $B$ remains in contact with the flat surface and on the $x$-axis.

The function $\mathrm{h}(t)$ satisfies the differential equation

$$\frac{\mathrm{d}^2\mathrm{h}}{\mathrm{d}t^2} = -\omega^2\mathrm{h}, \quad \text{with } \mathrm{h}(0) = 0 \text{ and } \frac{\mathrm{dh}}{\mathrm{d}t} = \omega L \text{ at } t = 0,$$

where $\omega$ is a positive constant.

A particle $P$ of mass $m$ remains in contact with the plank at point $C$.

**(i)** Show that the $x$-coordinate of $P$ is $sL\cos\omega t$, and find a similar expression for its $y$-coordinate.

**(ii)** Find expressions for the $x$- and $y$-components of the acceleration of the particle.

**(iii)** $N$ and $F$ are the upward normal and frictional components, respectively, of the force of the plank on the particle. Show that

$$N = mg(1 - k\sin\omega t)\cos\omega t,$$

and that

$$F = mgsk + N\tan\omega t$$

where $k = \dfrac{L\omega^2}{g}$.

**(iv)** The coefficient of friction between the particle and the plank is $\tan\alpha$, where $\alpha$ is an acute angle.

Show that the particle will not slip initially, provided $sk < \tan\alpha$.

Show further that, in this case, the particle will slip

- while $N$ is still positive,
- when the plank makes an angle less than $\alpha$ to the horizontal.

## Section C: Probability and Statistics

**11** **(i)** Let $\lambda > 0$. The independent random variables $X_1, X_2, \ldots, X_n$ all have probability density function

$$\mathrm{f}(t) = \begin{cases} \lambda \mathrm{e}^{-\lambda t} & t \geqslant 0 \\ 0 & t < 0 \end{cases}$$

and cumulative distribution function $\mathrm{F}(x)$.

The value of random variable $Y$ is the largest of the values $X_1, X_2, \ldots, X_n$. Show that the cumulative distribution function of $Y$ is given, for $y \geqslant 0$, by

$$\mathrm{G}(y) = \left(1 - \mathrm{e}^{-\lambda y}\right)^n .$$

**(ii)** The values $\mathrm{L}(\alpha)$ and $\mathrm{U}(\alpha)$, where $0 < \alpha \leqslant \frac{1}{2}$, are such that

$$\mathrm{P}(Y < \mathrm{L}(\alpha)) = \alpha \quad \text{and} \quad \mathrm{P}(Y > \mathrm{U}(\alpha)) = \alpha.$$

Show that

$$\mathrm{L}(\alpha) = -\frac{1}{\lambda} \ln\left(1 - \alpha^{\frac{1}{n}}\right)$$

and write down a similar expression for $\mathrm{U}(\alpha)$.

**(iii)** Use the approximation $\mathrm{e}^t \approx 1 + t$, for $|t|$ small, to show that, for sufficiently large $n$,

$$\lambda \mathrm{L}(\alpha) \approx \ln(n) - \ln\left(\ln\left(\frac{1}{\alpha}\right)\right).$$

**(iv)** Hence show that the median of $Y$ tends to infinity as $n$ increases, but that the width of the interval $\mathrm{U}(\alpha) - \mathrm{L}(\alpha)$ tends to a value which is independent of $n$.

**(v)** You are given that, for $|t|$ small, $\ln(1 + t) \approx t$ and that $\mathrm{e}^3 \approx 20$.

Show that, for sufficiently large $n$, there is an interval of width approximately $4\lambda^{-1}$ in which $Y$ lies with probability $0.9$.

**12** **(i)** Show that, for any functions f and g, and for any $m \geqslant 0$,

$$\sum_{r=1}^{m+1} \left( \mathrm{f}(r) \sum_{s=r-1}^{m} \mathrm{g}(s) \right) = \sum_{s=0}^{m} \left( \mathrm{g}(s) \sum_{r=1}^{s+1} \mathrm{f}(r) \right).$$

**(ii)** The random variables $X_0, X_1, X_2, \ldots$ are defined as follows

- $X_0$ takes the value 0 with probability 1;
- $X_{n+1}$ takes the values $0, 1, \ldots, X_n + 1$ with equal probability, for $n = 0, 1, \ldots$ .

**(a)** Write down $\mathrm{E}(X_1)$.

Find $\mathrm{P}(X_2 = 0)$ and $\mathrm{P}(X_2 = 1)$ and show that $\mathrm{P}(X_2 = 2) = \frac{1}{6}$.

Hence calculate $\mathrm{E}(X_2)$.

**(b)** For $n \geqslant 1$, show that

$$\mathrm{P}(X_n = 0) = \sum_{s=0}^{n-1} \frac{\mathrm{P}(X_{n-1} = s)}{s+2}$$

and find a similar expression for $\mathrm{P}(X_n = r)$, for $r = 1, 2, \ldots, n$.

**(c)** Hence show that $\mathrm{E}(X_n) = \frac{1}{2}(1 + \mathrm{E}(X_{n-1}))$.

Find an expression for $\mathrm{E}(X_n)$ in terms of $n$, for $n = 1, 2, \ldots$ .